



Launching VPC Resources

E

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VPC oluřtur

[Bilgi](#)

VPC, AWS Bulut'un Amazon EC2 bulut sunucuları gibi AWS nesneleri tarafından doldurulan yalıtılmıř bir bölümdür. İlgili kaynakları vurgulamak için fareyi bir kaynağın üzerine getirin.

VPC ayarları

Oluřturulacak kaynaklar

[Bilgi](#)

Yalnızca VPC kaynağının veya VPC ve diğeri ağ kaynaklarını oluřturun.

☐ Yalnızca VPC

☒ VPC ve daha fazlası

Otomatik ad etiketi oluřturma

[Bilgi](#)

Ad etiketi için bir değeri girin. Bu değeri, VPC'deki tüm kaynaklar için Ad etiketlerini otomatik olarak oluřturmak üzere kullanılır.

☒ Otomatik oluřturma

nextwork

IPv4 CIDR bloğuru

[Bilgi](#)

CIDR gösterimini kullanarak bağılangıç IP'sini ve VPC'nizin boyutunu belirleyin.

10.0.0.0/16

65.536 IP'si

CIDR blok boyutu /16 ile /28 arasında olmalıdır.

IPv6 CIDR bloğuru

[Bilgi](#)

☒ IPv6 CIDR bloğuru yok

☐ Amazon tarafından sağlanan IPv6 CIDR bloğuru

Ön izleme

VPC

[Ayrıntıları göster](#)

AWS sanal ağıdır

nextwork-vpc

Alt ağlar (4)

Bu VPC içindeki alt ağlar

eu-central-1a

nextwork-subnet-public1-eu-

nextwork-subnet-private1-eu-

eu-central-1b

nextwork-subnet-public2-eu-

nextwork-subnet-private2-eu-

Yönlendirme t

Ağ trafiğini kaynaklar:

nextwork-rtb-publ

nextwork-rtb-priv

nextwork-rtb-priv



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a logically isolated section of the AWS cloud where you can launch AWS resources in a virtual network that you define and control. It is essentially your own private data center within AWS.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to build a complete, multi-tier network architecture with the following components and configurations: Created a VPC with a custom IPv4 CIDR block (e.g., 10.0.0.0/16) to establish my isolated network boundary. Created public and private subnets across multiple Availability Zones for high availability and fault tolerance: Public subnets for resources that need direct internet access (web servers, bastion hosts). Private subnets for resources that should be protected from direct internet exposure (databases, application servers). Created an Internet Gateway (IGW) and attached it to the VPC to enable internet connectivity for public subnets. Created route tables and associated them with subnets: Public route table: route 0.0.0.0/0 → Internet Gateway. Private route table: route 0.0.0.0/0 → NAT Gateway (for outbound-only internet access).

One thing I didn't expect in this project was...

One thing I didn't expect was how tightly interconnected all the VPC components are, and how a single missing association can break the entire network flow.



This project took me...

This project took me approximately 2.5 hours to complete.



Setting Up Direct VM Access

To directly access your EC2 instance means to connect to it from your local machine (or another source) without going through intermediate servers or proxies. For Linux instances, this typically means using SSH (Secure Shell) from your computer's terminal. For Windows instances, it means using RDP (Remote Desktop Protocol). Direct access gives you full control over the operating system, allowing you to run commands, install software, and manage files as if you were sitting in front of the actual server.

SSH is a key method for directly accessing a VM

SSH stands for Secure Shell. It is a cryptographic network protocol used to securely connect to and manage remote servers over an unsecured network. SSH provides: Encrypted communication — all data between your local machine and the server is encrypted. Authentication — verifies your identity using key pairs or passwords. Remote command execution — allows you to run commands on the remote server as if you were physically there. In AWS, SSH is the standard way to access Linux EC2 instances (typically on port 22).

To enable direct access, I set up key pairs

A key pair consists of two cryptographic keys: a public key and a private key. AWS stores the public key on your EC2 instance, while you download and securely store the private key. It is used to prove your identity when connecting to your EC2 instance via SSH (Secure Shell). Think of it as a more secure alternative to passwords — instead of typing a password, your private key authenticates you automatically.

A private key's file format means the specific file extension and encoding used to store the cryptographic key. The format determines how the key is structured and which tools can use it. My private key's file format was .pem (Privacy Enhanced Mail format).



This is the default format AWS provides when you create a key pair. It is a base64-encoded ASCII file containing the private key. On Windows, you might convert this to .ppk format (PuTTY Private Key) if using PuTTY as your SSH client.



Launching a public server

I had to change my EC2 instance's networking settings by modifying the following:

- Subnet assignment — Ensuring the EC2 instance was placed in the correct subnet (public subnet for internet-facing instances, private subnet for internal-only resources).
- Security Group rules — Configuring inbound rules to allow SSH (port 22) or HTTP/HTTPS traffic from specific IP addresses, and outbound rules to allow internet access.
- Network ACLs — Adjusting subnet-level firewall rules to permit necessary traffic flows.
- Route Table associations — Ensuring the subnet's route table directs traffic properly (e.g., 0.0.0.0/0 → Internet Gateway for public subnets).
- Auto-assign Public IP — Enabling or disabling automatic public IP assignment depending on whether the instance needs direct internet accessibility.
- Elastic IP (optional) — Associating a static public IP address if the instance needs a persistent, unchanging IP for external access.

The screenshot displays the AWS Management Console for an EC2 instance named 'i-072d542f23fbe5894' (NextWork Public Server). The instance is running in the 'eu-central-1a' availability zone. The 'Ağ iletişimi' (Networking) tab is selected, showing the following configuration:

- VPC kimliği:** vpc-0b728ac972c5d2909 (NextWork VPC)
- Alt ağ kimliği:** subnet-0e1460d36a90b2dec (NextWork Public Subnet)
- Erişilebilirlik alanı:** eu-central-1a
- Erişilebilirlik alanı ID'si:** euc1-az2
- Outpost kimliği:** -

Below the configuration, there are sections for IP addresses:

- IP adresleri:** A dropdown menu showing 'Genel IPv4 adresi' (Public IPv4 address).
- Özel IPv4 adresleri:** A section for private IPv4 addresses.
- IPv6 adresleri:** A section for IPv6 addresses.



Launching a private server

By using separate security groups: You prevent accidental misconfiguration that could expose private resources. You enforce defense in depth — even if the public server is compromised, the private server has its own independent firewall. You follow AWS best practices — each tier (public web tier, private app/database tier) should have its own security group tailored to its specific role.

Specifically, for your private server's security group: SSH (port 22): Source = Security Group ID of your public EC2 instance (e.g., sg-0abcd1234) — this allows SSH access ONLY from instances that have that specific security group attached. Alternatively: Source = Your VPC's CIDR block (e.g., 10.0.0.0/16) or the public subnet's CIDR (e.g., 10.0.1.0/24) — this allows traffic from any resource within those IP ranges. Key point: Using a security group ID as a source (rather than an IP address) is the most secure and flexible approach. It means that any EC2 instance with that security group can automatically communicate with your private instance, regardless of its specific IP address.

Devre dışı bırak

Güvenlik duvarı (güvenlik grupları) | [Bilgi](#)
Bir güvenlik grubu, bulut sunucunuzun trafiğini kontrol eden bir dizi güvenlik duvarı kuralıdır. Belirli trafiğin bulut sunucunuza ulaşmasına izin vermek için kurallar ekleyin.

☒ Güvenlik grubu oluştur ☐ Mevcut güvenlik grubunu seç

Güvenlik grubu adı - *gerekli*
NextWork Private Security Group
Bu güvenlik grubu tüm ağ arabirimlerine eklenecektir. Güvenlik grubu oluşturulduktan sonra ad düzenlenemez. Maksimum uzunluk 255 karakterdir. Geçerli karakterler: a-z, A-Z, 0-9, boşluklar ve _-~/() #, @ [] +=&; {}! \$*

Açıklama - *gerekli* | [Bilgi](#)
Security group for NextWork Private Subnet.

Gelen Güvenlik Grubu Kuralları
▼ Güvenlik grubu kuralı 1 (TCP, 22, sg-0e373fdde788e74ee) [Kaldır](#)

Türü | [Bilgi](#)
ssh

Protokol | [Bilgi](#)
TCP

Bağlantı noktası aralığı | [Bilgi](#)
22

Kaynak türü | [Bilgi](#)
Özel

Kaynak | [Bilgi](#)
Q CIDR, ön ek listesi veya güvenlik grubu e
sg-0e373fdde788e74ee

Açıklama - *isteğe bağlı* | [Bilgi](#)
Ör. yönetici masaüstü için SSH



Speeding up VPC creation

I set up my new VPC using the VPC creation wizard (or VPC resource map / "VPC and more" option in the AWS Console), which allows you to: Define the VPC CIDR block (e.g., 10.0.0.0/16) Specify the number of Availability Zones (e.g., 2 AZs for high availability) Choose public and private subnets per AZ Automatically create NAT gateways (one per AZ or single NAT gateway) Auto-generate route tables with proper associations Automatically create and attach an Internet Gateway Set up VPC endpoints (optional)

A VPC resource map is a visual diagram in the AWS Console that shows the complete topology of your VPC infrastructure. It displays: Your VPC and its CIDR range All subnets (public and private) with their CIDR blocks and AZs Route tables associated with each subnet Internet Gateway and NAT Gateway connections VPC endpoints (if any) Network ACLs and Security Groups It acts as a single-pane-of-glass dashboard where you can see how all resources are interconnected, troubleshoot connectivity issues, and verify your architecture matches your design intent. You can access it in the AWS Console under VPC → Your VPCs → Resource map tab.

Your new VPC can have the same IPv4 CIDR block (e.g., 10.0.0.0/16) as the NextWork VPC because VPC CIDR blocks are private and isolated per VPC. AWS VPCs operate in a private, isolated network space — the IP addresses you choose are only meaningful within your AWS account and within that specific VPC. Two VPCs in the same AWS account can technically have overlapping CIDR blocks, but they cannot be peered or connected without causing routing conflicts. However, for learning purposes or if the VPCs are completely separate (no peering, no VPN, no Transit Gateway), using the same CIDR is allowed because they are network-isolated from each other. Important caveat: If you later need to connect these VPCs (via VPC peering, VPN, or AWS Transit Gateway), overlapping CIDR blocks will cause conflicts and must be avoided. In production, always use unique, non-overlapping CIDR ranges.



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AWS sanal ağıdır

proje-vpc

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eu-central-1b

G proje-subnet-public2-eu-central-1b

B proje-subnet-private2-eu-central-

Yönlendirme tabloları (3)

Ağ trafiğini kaynaklara yönlendir

proje-rtb-public

proje-rtb-private1-eu-central-1a

proje-rtb-private2-eu-central-1b



Speeding up VPC creation

Tips for using the VPC resource map

You could create up to the maximum number of subnets allowed per VPC, which is 200 subnets per VPC (this is a soft limit that can be increased by requesting a quota increase from AWS). However, practically speaking: You typically create one public subnet per Availability Zone (e.g., 2 AZs = 2 public subnets, 3 AZs = 3 public subnets) for high availability and fault tolerance. You could create more public subnets if you have specific needs (e.g., separate public subnets for different application tiers, DMZ subnets, or reserved ranges for future expansion). Each subnet must have a unique, non-overlapping CIDR block within the VPC's overall CIDR range. Example: In a 10.0.0.0/16 VPC, you could create 256 /24 subnets (each with 256 IPs), mixing public and private as needed.

NAT gateways (Network Address Translation gateways) are AWS-managed services that allow instances in private subnets to initiate outbound internet connections (for software updates, API calls, package downloads) while preventing inbound internet connections from reaching those instances.



VPC oluřtur [Bilgi](#)

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4 nextwork-subnet-private2-eu-

Yönlendirme t

Ağı trafiğini kaynaklar

nextwork-rtb-publ

nextwork-rtb-priv

nextwork-rtb-priv



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